

RIALS FOR ACOUSTICS: METAMATERIALS, SONIC CRYSTALS AND CALCULATED MICROSTRUCTURES

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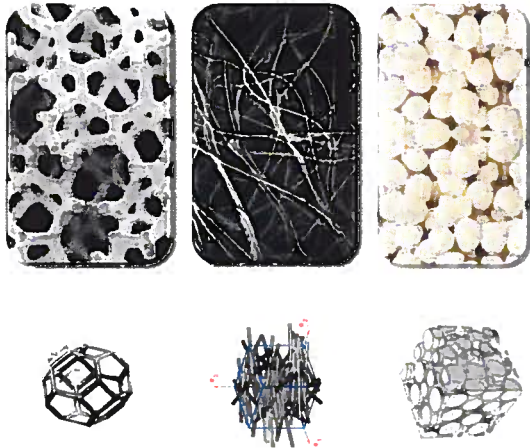


FIGURE 1 – Examples of conventionnal materials

2.2 Engineered materials for acoustics (EMA)

It may sound a little pompous, but while nature aims for energy minima in the manufacture of materials, Man aims for maximums in the performance of materials. Thus, by mastering the physics behind wave propagation and dissipation in air-saturated media, one finds complex optimal geometries that maximize, for example, sound absorption or transmission loss in a given frequency range. This is why such materials are called engineered materials or structured materials. The structuring of the materials may be at different scales : microscopic (much smaller than wavelength), mesoscopic (smaller than wavelength), and macroscopic (comparable to wavelength).

One of the first types of macroscopically structured acoustic materials was non-resonant sonic crystals (SC). A SC is a periodic arrangement of scatterers. They present band gap in the transmission of sound. The central frequency of the first band gap is given by the Bragg condition : $\lambda/H = 2$,

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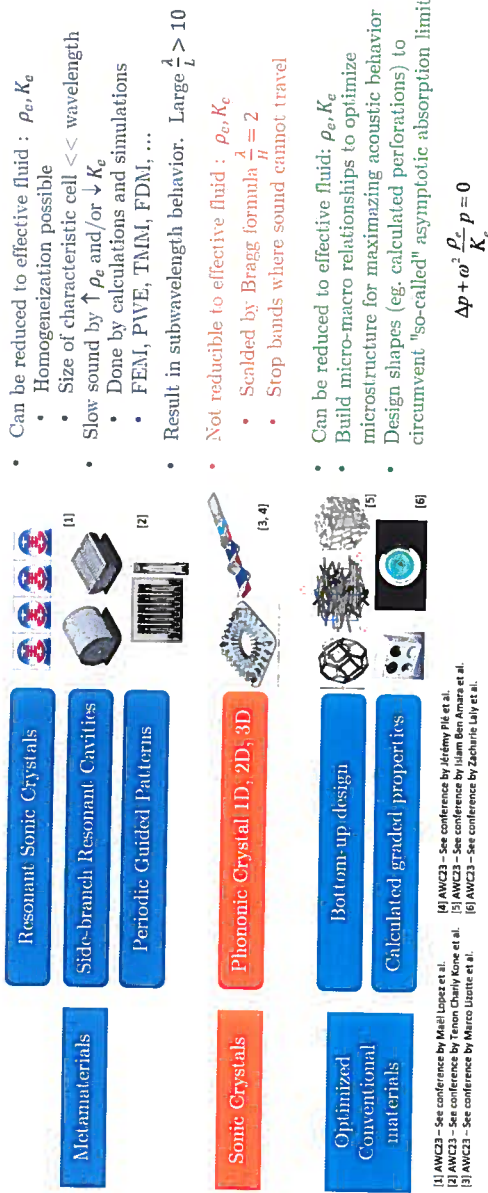


FIGURE 2 – Classification of engineered materials for acoustics.

they are also called sub-wavelength or slow sound materials. Much of the articles in the literature refer to these materials as metamaterials. Therefore, metamaterials shift the acoustic spectrum to lower frequencies and can achieve a wavelength-to-thickness ratio well over 10. Additionally, due to the resonances, their types and periodic arrangement, the wavenumber may be either positive real, imaginary, or negative real. This has the effect of creating band gaps of different natures.

Finally, at the microscale, one can find materials for which their microstructure and transport parameters are optimized by calculations to maximize thermoviscous dissipation. The author calls these materials optimized conventional materials. On the one hand, they may be conventional materials whose microstructure has been optimized by multiscale calculations (bottom-up approach). For example, sheets of fibers whose angular orientation and polydispersity of fiber diameters have been optimized to obtain the best acoustic absorption at low frequencies. In these cases, wavelength-to-thickness ratios as high as 10, even 20, can be achieved, which is more than twice the ratio of conventional materials. On the other hand, porous materials with graded properties can also be considered. The gradation can be continuous or stepwise.

3 Classification

As described in the previous paragraphs, three distinct classes emerge : sonic crystals (SC), metamaterials (MM), and optimized conventional materials (OCM).

the mesoscopic and microscopic scales, they can also be homogenized at the macroscopic scale (wavelength scale) as for classical acoustic materials. Therefore, OCM and MM can be represented as equivalent fluids of effective properties for which the Helmholtz equation governs sound propagation.

Based on the previous short definitions, Figure 2 presents the classification proposed by the author. This classification and the underlying definitions are not the result of a consensus of the scientific community in acoustics. The author proposes here a classification that seems consistent with what is found in the literature. It remains to be debated and validated by peers. Its presentation at this conference is only one step.

4 Discussion and conclusion

As shown in Figure 2, the engineered materials for acoustics are divided in three classes : MM, SC and OCM. Note that reality is more complex than this simple classification. Indeed, acoustic solutions may imply combinations of many classes. For instance, a combination of SC with resonant MM elements, such as resonant sonic crystals, or a combination of side-branch resonant cavities embedded in a conventional porous materials. In this classification, the most important thing to keep in mind is the description given on the right side of the figure, or the wavelength-to-thickness ratio that each class can achieve. Based on this, an intelligent decision on the type of noise control elements for a given application can be made